

**AN EXPLICIT USL MATERIAL POINT
METHOD
IMPLEMENTATION OF
VARIATIONAL PHASE–FIELD FRACTURE**

Technical Report

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ATHENS, JUNE 2026

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1 Introduction and Scope

This report develops an *explicit* solution scheme for variational phase-field fracture, formulated within the *Update–Stress–Last* (USL) Material Point Method (MPM). The continuum phase-field theory is adopted essentially unchanged from the variational fracture framework summarised below, and the same symbols are retained throughout: the displacement field $\mathbf{u}$, the scalar phase field (damage variable) $\phi \in [0, 1]$, the degradation function $g(\phi)$, the crack surface density function $\gamma(\phi, \nabla\phi)$, the fracture toughness G_c , the length scale ℓ , the normalisation constant c_w , the local crack geometry function $w(\phi)$, the (undamaged) elastic strain energy density ψ^e , and the history field $\mathcal{H}$.

The original framework is intrinsically *quasi-static*: the scalar micro-stress ω that is work conjugate to ϕ is purely *energetic*, so the phase-field balance $\nabla \cdot \boldsymbol{\xi} - \omega = 0$ is an *elliptic* equation that must be solved by an implicit (monolithic or staggered) Newton iteration at every time step. This is incompatible with an explicit MPM time integrator, in which all unknowns are marched forward in time without solving a linear system.

The central modification of this report is the introduction of a *dissipative* contribution to the phase-field micro-stress,

$$\omega_{\text{diss}} = \eta \dot{\phi} = \eta \frac{d\phi}{dt}, \quad \eta \geq 0, \quad (1)$$

where η is a (artificial) viscosity with units of $\text{Pa} \cdot \text{s}$. As shown by Kakouris & Triantafyllou and by Cheon & Kim, this single term turns the damage sub-problem from an *elliptic* into a *parabolic* (Allen–Cahn type) equation, so the phase field can be advanced in time by an explicit forward–Euler update on a lumped pseudo-mass, exactly in the spirit of the explicit momentum update of the displacement problem. The result is a fully explicit, monolithic-in-time USL–MPM algorithm in which displacement and phase field are updated within the *same* time loop.

The report is organised as follows. Section 2 recalls the phase-field regularisation of Griffith fracture. Section 3 restates the principle of virtual work, now augmented with the dissipative micro-force (1), and derives the strong forms. Section 4 gives the constitutive theory and the AT1/AT2 crack density choices. Section 5 collects the coupled strong forms, emphasising the parabolic nature of the regularised evolution law. Section 6 treats damage irreversibility, the weak form and the MPM space discretisation, and derives the *explicit nodal phase-field update*. Finally, Section 7 embeds this update into the explicit USL–MPM cycle and states the complete solution algorithm, followed by stability and time-step considerations.

2 Phase-field Regularisation of Griffith Fracture

Consider a solid of volume V occupying a domain $\Omega \subset \mathbb{R}^n$ ($n \in \{1, 2, 3\}$) with external boundary $\partial\Omega$ and outward unit normal $\mathbf{n}$. The elastic strain energy density of the undamaged solid is $\psi^e(\boldsymbol{\varepsilon})$, a function of the (small) strain tensor $\boldsymbol{\varepsilon}$.

Griffith’s energy balance states that the variation of the total energy Π with respect to an incre-

mental increase of the crack area dA vanishes,

$$\frac{d\Pi}{dA} = \frac{d\psi^e(\boldsymbol{\varepsilon})}{dA} + \frac{dW_c}{dA} = 0, \quad (2)$$

where W_c is the work required to create new surfaces and $G_c = dW_c/dA$ is the critical energy release rate. In variational form,

$$\Pi = \int_{\Omega} \psi^e(\boldsymbol{\varepsilon}) dV + \int_{\Gamma} G_c d\Gamma, \quad (3)$$

with Γ the (unknown) crack surface. The difficulty posed by the unknown Γ is removed by introducing an auxiliary phase field ϕ that smears the sharp interface into a diffuse region: $\phi = 0$ in intact material and $\phi = 1$ in fully cracked material. The Griffith functional is then approximated by the regularised functional

$$\Pi_{\ell} = \int_{\Omega} [g(\phi) \psi_0^e(\boldsymbol{\varepsilon}) + G_c \gamma(\phi, \ell)] dV, \quad (4)$$

where ℓ is a length scale governing the size of the fracture process zone, ψ_0^e is the elastic energy density of the undamaged solid, and $\gamma(\phi, \ell)$ is the crack surface density function. The degradation function $g(\phi)$ degrades the stiffness as $\phi \rightarrow 1$. As $\ell \rightarrow 0^+$ the functional Π_{ℓ} Γ -converges to the sharp Griffith functional (3); for $\ell > 0^+$ a finite material strength is introduced, so that ℓ becomes a material property governing strength, e.g. for plane stress

$$\sigma_c \propto \sqrt{\frac{G_c E}{\ell}} = \frac{K_{Ic}}{\sqrt{\ell}}, \quad (5)$$

with K_{Ic} the fracture toughness and E Young's modulus.

3 Principle of Virtual Work and Dissipative Micro-force

The primal kinematic variables are the displacement field $\mathbf{u}$ and the phase field ϕ . Restricting attention to small strains and isothermal conditions, the strain tensor is

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u}^T + \nabla \mathbf{u}). \quad (6)$$

Let δ denote virtual quantities. The Cauchy stress $\boldsymbol{\sigma}$ is work conjugate to $\boldsymbol{\varepsilon}$, the traction $\mathbf{T}$ is work conjugate to $\mathbf{u}$, the scalar stress-like quantity ω is work conjugate to ϕ , and the phase-field micro-stress vector $\boldsymbol{\xi}$ is work conjugate to $\nabla \phi$. Because the phase field is driven by the displacement problem alone, no external traction is associated with ϕ . In the absence of body forces the principle of virtual work reads

$$\int_{\Omega} \{\boldsymbol{\sigma} : \delta \boldsymbol{\varepsilon} + \omega \delta \phi + \boldsymbol{\xi} \cdot \nabla \delta \phi\} dV = \int_{\partial \Omega} (\mathbf{T} \cdot \delta \mathbf{u}) dS. \quad (7)$$

Energetic and dissipative parts of the micro-stress. The key step for an explicit scheme is to allow the phase-field micro-stress ω to possess *both* an energetic and a dissipative part,

$$\omega = \omega_{\text{en}} + \omega_{\text{diss}}, \quad \omega_{\text{en}} = \frac{\partial \psi}{\partial \phi}, \quad \omega_{\text{diss}} = \eta \dot{\phi}, \quad (8)$$

where ψ is the free energy density (defined in Section 4) and $\eta \geq 0$ is the phase-field viscosity. The dissipative part $\omega_{\text{diss}} = \eta \dot{\phi}$ is the minimal rate-dependent contribution that satisfies the dissipation inequality $\omega_{\text{diss}} \dot{\phi} = \eta \dot{\phi}^2 \geq 0$ and vanishes in the rate-independent (quasi-static) limit $\eta \rightarrow 0$. The micro-stress vector ξ is taken purely energetic, $\xi = \partial \psi / \partial \nabla \phi$.

Local balance laws (strong form). Since (7) must hold for an arbitrary domain Ω and for any kinematically admissible variation, the fundamental lemma of the calculus of variations yields the local force balances

$$\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}, \quad \nabla \cdot \boldsymbol{\xi} - \omega = 0 \quad \text{in } \Omega, \quad (9)$$

with natural boundary conditions

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{T}, \quad \boldsymbol{\xi} \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega. \quad (10)$$

Substituting the split (8) into the phase-field balance $\nabla \cdot \boldsymbol{\xi} - \omega = 0$ gives the *rate* form

$$\boxed{\eta \dot{\phi} = \nabla \cdot \boldsymbol{\xi} - \omega_{\text{en}}.} \quad (11)$$

Equation (11) is the cornerstone of the explicit scheme: for $\eta > 0$ it is a first-order-in-time evolution law that can be integrated by forward Euler, whereas for $\eta = 0$ it collapses to the original elliptic stationarity condition $\nabla \cdot \boldsymbol{\xi} - \omega_{\text{en}} = 0$. The dynamic momentum balance used by the explicit MPM replaces the first of (9) by $\rho \dot{\mathbf{v}} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{b}$, with ρ the density and $\mathbf{b}$ the body force per unit mass.

4 Constitutive Theory

The free energy density of the solid is additively split into an elastic (bulk) part and a fracture part,

$$\psi(\boldsymbol{\varepsilon}, \phi, \nabla \phi) = \psi^e + \psi^f = g(\phi) \frac{1}{2} \boldsymbol{\varepsilon} : \mathbf{C}_0 : \boldsymbol{\varepsilon} + G_c \gamma(\phi, \nabla \phi), \quad (12)$$

where $\mathbf{C}_0$ is the linear elastic stiffness tensor and $\psi_0^e = \frac{1}{2} \boldsymbol{\varepsilon} : \mathbf{C}_0 : \boldsymbol{\varepsilon}$ is the undamaged elastic energy density. The Cauchy stress follows as

$$\boldsymbol{\sigma} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} = g(\phi) (\mathbf{C}_0 : \boldsymbol{\varepsilon}) = g(\phi) \boldsymbol{\sigma}_0, \quad \boldsymbol{\sigma}_0 \equiv \mathbf{C}_0 : \boldsymbol{\varepsilon}, \quad (13)$$

with $\boldsymbol{\sigma}_0$ the effective (undamaged) stress.

Degradation function. The degradation function is continuous and monotonic with $g(0) = 1$ and $g(1) = 0$; the standard quadratic form is adopted,

$$g(\phi) = (1 - \phi)^2, \quad g'(\phi) = -2(1 - \phi), \quad g''(\phi) = 2. \quad (14)$$

Crack surface density (AT1 / AT2). Following the Ambrosio–Tortorelli family, the crack surface density function and the crack surface A are

$$A = \int_{\Omega} \gamma(\phi) dV = \int_{\Omega} \frac{1}{4c_w \ell} \left(w(\phi) + \ell^2 |\nabla \phi|^2 \right) dV, \quad (15)$$

where the local crack geometry function $w(\phi)$ satisfies $w(0) = 0$, $w(1) = 1$, and the normalisation constant is

$$c_w = \int_0^1 \sqrt{w(\varphi)} d\varphi. \quad (16)$$

Two standard choices are retained throughout this report:

$$\textbf{AT2:} \quad w(\phi) = \phi^2, \quad c_w = \frac{1}{2}, \quad w'(\phi) = 2\phi, \quad w''(\phi) = 2, \quad (17)$$

$$\textbf{AT1:} \quad w(\phi) = \phi, \quad c_w = \frac{2}{3}, \quad w'(\phi) = 1, \quad w''(\phi) = 0. \quad (18)$$

The AT2 model degrades from the onset of loading, whereas the AT1 model exhibits a purely elastic regime prior to damage onset. The critical (homogeneous, $\nabla \phi = 0$) failure stresses are

$$\sigma_c^{\text{AT1}} = \sqrt{\frac{3EG_c}{8\ell}}, \quad \sigma_c^{\text{AT2}} = \frac{3}{16} \sqrt{\frac{3EG_c}{\ell}}. \quad (19)$$

Micro–stresses. From (12), the energetic scalar micro–stress and the micro–stress vector are

$$\omega_{\text{en}} = \frac{\partial \psi}{\partial \phi} = g'(\phi) \psi^e + \frac{1}{4c_w \ell} G_c w'(\phi), \quad (20)$$

$$\xi = \frac{\partial \psi}{\partial \nabla \phi} = \frac{\ell}{2c_w} G_c \nabla \phi. \quad (21)$$

Together with the dissipative part (8), the total micro–stress is $\omega = g'(\phi) \psi^e + \frac{1}{4c_w \ell} G_c w'(\phi) + \eta \dot{\phi}$.

5 Coupled Strong Forms

Collecting (11), (20) and (21), the regularised phase–field evolution law becomes the parabolic (Allen–Cahn type) equation

$$\boxed{\eta \dot{\phi} = \frac{\ell}{2c_w} G_c \nabla^2 \phi - \frac{G_c}{4c_w \ell} w'(\phi) - g'(\phi) \psi^e(\epsilon)}, \quad (22)$$

obtained from $\nabla \cdot \xi = \frac{\ell}{2c_w} G_c \nabla^2 \phi$. Equivalently, using $g'(\phi) = -2(1 - \phi)$, the right–hand side expresses the competition between the stored elastic energy (which drives $\phi \rightarrow 1$) and the fracture energy (which penalises crack surface). In the quasi–static limit $\eta \rightarrow 0$, (22) reduces to the stationarity condition of the original theory,

$$\frac{G_c}{2c_w \ell} \left(\frac{w'(\phi)}{2} - \ell^2 \nabla^2 \phi \right) + g'(\phi) \psi^e(\epsilon) = 0. \quad (23)$$

The complete coupled boundary/initial–value problem to be solved is therefore

$$\rho \dot{\mathbf{v}} = \nabla \cdot [g(\phi) \boldsymbol{\sigma}_0] + \rho \mathbf{b}, \quad (\text{momentum}) \quad (24a)$$

$$\eta \dot{\phi} = \frac{\ell}{2c_w} G_c \nabla^2 \phi - \frac{G_c}{4c_w \ell} w'(\phi) - g'(\phi) \mathcal{H}, \quad (\text{phase field}) \quad (24b)$$

$$\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{T}, \quad \boldsymbol{\xi} \cdot \mathbf{n} = \nabla \phi \cdot \mathbf{n} = 0, \quad (\text{boundary}) \quad (24c)$$

where, anticipating Section 6, the elastic energy ψ^e driving the phase field has been replaced by the irreversible history field $\mathcal{H}$. The two equations are coupled through $g(\phi)$ (the phase field degrades the stress in the momentum balance) and through $\psi^e/\mathcal{H}$ (the elastic energy drives the phase field).

6 Discretisation and Explicit Phase–field Update

6.1 Damage irreversibility and history field

To prevent crack healing, the phase field must be monotonically non–decreasing,

$$\phi_{t+\Delta t} \geq \phi_t. \quad (25)$$

Following Miehe *et al.*, this is enforced by replacing the instantaneous elastic energy ψ_0^e with a history field $\mathcal{H}$ that records the maximum elastic energy density attained over the loading history,

$$\mathcal{H} = \max_{\tau \in [0, t]} \psi_0^e(\tau), \quad \psi_0^e - \mathcal{H} \leq 0, \quad \dot{\mathcal{H}} \geq 0, \quad \dot{\mathcal{H}} (\psi_0^e - \mathcal{H}) = 0. \quad (26)$$

The crack driving force in the phase–field equation is thus $\mathcal{H}$, which is updated particle–wise (Section 7) and is monotone by construction.

6.2 Weak form of the damage sub–problem

Adopting a total–Lagrangian description for the damage equation and using the dissipative micro–force (8), the weak form of the phase–field balance (11) over the reference domain Ω_0 reads, for any admissible test function $\delta\phi$,

$$\int_{\Omega_0} \left\{ g'(\phi) \mathcal{H} \delta\phi + \frac{G_c}{2c_w \ell} \left[\frac{1}{2} w'(\phi) \delta\phi + \ell^2 \nabla_0 \phi \cdot \nabla_0 \delta\phi \right] + \eta \dot{\phi} \delta\phi \right\} dV = 0. \quad (27)$$

The first three terms are exactly the energetic residual of the original quasi–static formulation; the last term, $\eta \dot{\phi} \delta\phi$, is the new dissipative contribution that renders the problem parabolic.

6.3 Spatial discretisation

Both the displacement and the phase field are interpolated with the same shape functions. In the MPM the shape functions are the *background grid* functions N_I (equivalently ϕ_I in some

MPM notations); we write N_I to avoid clashing with the phase field ϕ . Nodal quantities are interpolated as

$$\mathbf{u} = \sum_I N_I \hat{\mathbf{u}}_I, \quad \phi = \sum_I N_I \hat{\phi}_I, \quad \boldsymbol{\varepsilon} = \sum_I \mathbf{B}_I^u \hat{\mathbf{u}}_I, \quad \nabla \phi = \sum_I \mathbf{B}_I \hat{\phi}_I, \quad (28)$$

with $\mathbf{B}_I^u$ the strain–displacement matrix and $\mathbf{B}_I = \nabla N_I$ the gradient of the nodal shape function. Inserting (28) into the weak form (27) yields, at node I , the residual

$$R_I^\phi = \int_{\Omega_0} \left\{ g'(\phi) N_I \mathcal{H} + \frac{G_c}{2c_w \ell} \left[\frac{1}{2} w'(\phi) N_I + \ell^2 \mathbf{B}_I^T \nabla_0 \phi \right] + \eta \dot{\phi} N_I \right\} dV = 0. \quad (29)$$

For reference, the (small–strain) displacement residual and the consistent tangents of the implicit formulation are

$$\mathbf{R}_I^u = \int_{\Omega_0} [g(\phi) + \kappa] (\mathbf{B}_I^u)^T \boldsymbol{\sigma}_0 dV, \quad (30)$$

$$\mathbf{K}_{IJ}^u = \int_{\Omega_0} [g(\phi) + \kappa] (\mathbf{B}_I^u)^T \mathbf{C}_0 \mathbf{B}_J^u dV, \quad (31)$$

$$K_{IJ}^\phi = \int_{\Omega_0} \left\{ \left(g''(\phi) \mathcal{H} + \frac{G_c}{4c_w \ell} w''(\phi) \right) N_I N_J + \frac{G_c \ell}{2c_w} \mathbf{B}_I^T \mathbf{B}_J \right\} dV, \quad (32)$$

where $\kappa \approx 10^{-7}$ is a small residual stiffness that prevents ill–conditioning when $\phi = 1$. The explicit scheme below does *not* assemble or invert K^ϕ ; it is reproduced only to make the connection with the implicit formulation explicit.

6.4 Explicit nodal phase–field update

Discretising the dissipative term in (29) with a forward (explicit) Euler approximation of the rate,

$$\dot{\phi} \approx \frac{\phi_I^{t+\Delta t} - \phi_I^t}{\Delta t}, \quad (33)$$

and lumping(eg row-sum rule) the viscous term onto a diagonal *pseudo–mass*

$$m_I := \int_{\Omega_0} N_I dV, \quad (34)$$

the nodal balance $R_I^\phi = 0$ can be solved *explicitly* for the updated nodal phase field. Writing the energetic part of the residual as F_I ,

$$F_I = \int_{\Omega_0} \left\{ g'(\phi^t) N_I \mathcal{H} + \frac{G_c}{2c_w \ell} \left[\frac{1}{2} w'(\phi^t) N_I + \ell^2 \mathbf{B}_I^T \nabla_0 \phi^t \right] \right\} dV, \quad (35)$$

the update reads

$$\boxed{\phi_I^{t+\Delta t} = \phi_I^t - \frac{\Delta t}{\eta m_I} F_I.} \quad (36)$$

MPM (particle) quadrature. In the MPM the reference–domain integrals are evaluated using the material points as quadrature points, each carrying a reference volume V_p^0 , so that $\int_{\Omega_0}(\cdot) dV \approx \sum_p(\cdot)_p V_p^0$ and the pseudo–mass becomes $m_I = \sum_p N_I(\mathbf{x}_p) V_p^0$. The explicit nodal update in fully discrete MPM form is

$$\phi_I^{t+\Delta t} = \phi_I^t - \frac{\Delta t}{\eta m_I} \sum_p \left[g'(\phi_p^t) N_I(\mathbf{x}_p) \mathcal{H}_p + \frac{G_c}{2c_w \ell} \left(\frac{1}{2} w'(\phi_p^t) N_I(\mathbf{x}_p) + \ell^2 \nabla N_I(\mathbf{x}_p) \cdot \nabla \phi^t \right) \right] V_p^0. \quad (37)$$

Here $\phi_p^t = \sum_I N_I(\mathbf{x}_p) \phi_I^t$ and $\nabla \phi^t = \sum_I \nabla N_I(\mathbf{x}_p) \phi_I^t$ are the phase field and its gradient evaluated at the particle, and $\mathcal{H}_p$ is the particle history field. After the grid update (37), the particle phase field is recovered by mapping back from the grid,

$$\phi_p^{t+\Delta t} = \sum_I N_I(\mathbf{x}_p) \phi_I^{t+\Delta t}, \quad (38)$$

and the irreversibility constraint (25) is enforced point–wise, $\phi_p^{t+\Delta t} \leftarrow \max(\phi_p^t, \phi_p^{t+\Delta t})$.

Remark (mapping with the original symbols). The structure of (37) is identical to the explicit ULMPM–PFM update of Kakouris & Triantafyllou / Cheon & Kim, under the symbol correspondence $d \leftrightarrow \phi$, $\omega(d) \leftrightarrow g(\phi)$, $\check{Y} \leftrightarrow \mathcal{H}$, $\alpha(d) \leftrightarrow w(\phi)$, $b \leftrightarrow \ell$, $c_\alpha \leftrightarrow c_w$ (up to the chosen normalisation of γ), and the viscosity $\xi \leftrightarrow \eta$. We retain the symbols of the variational theory throughout.

7 Total Solution Algorithm: Explicit USL–MPM with Phase Field

The phase–field update (37) is now embedded in the explicit USL (Update–Stress–Last) MPM cycle. In USL, the particle stresses are updated *last*, i.e. after the grid momenta have been advanced and mapped back to the particles. Each material point p carries position $\mathbf{x}_p$, velocity $\mathbf{v}_p$, stress $\boldsymbol{\sigma}_p$ (or deformation gradient $\mathbf{F}_p$), volumes V_p^0, V_p , mass m_p , density ρ_p , phase field ϕ_p and history $\mathcal{H}_p$.

The phase field is advanced within the same particle–to–grid / grid–to–particle machinery as the momentum problem:

1. In the *particle–to–grid* (P2G) sweep, the pseudo–mass $m_I = \sum_p N_I(\mathbf{x}_p) V_p^0$ and the energetic damage force F_I of (35) are accumulated, alongside the mechanical nodal mass, momentum and forces.
2. The grid phase field is advanced explicitly by (37), in lock–step with the explicit momentum update.
3. In the *grid–to–particle* (G2P) sweep, the particle phase field is updated by (38), the history field $\mathcal{H}_p$ is refreshed from the current elastic energy, and the particle stresses are updated last using the degraded constitutive law $\boldsymbol{\sigma}_p = g(\phi_p) \boldsymbol{\sigma}_{0,p}$.

The complete procedure is given in Algorithm 1. Lines that are new relative to the standard USL–MPM (Algorithm 1 of the reference formulation) are those handling the phase field: the pseudo–mass and damage force accumulation in the P2G sweep, the explicit grid damage update, and the particle damage / history update in the G2P sweep.

Algorithm 1 Solution procedure of explicit USL–MPM with phase–field fracture

- 1: **Initialisation:** set up Cartesian grid; $t \leftarrow 0$
 - 2: Set up particle data: $\mathbf{x}_p^0, \mathbf{v}_p^0, \boldsymbol{\sigma}_p^0, \mathbf{F}_p^0, V_p^0, m_p, \rho_p^0, \phi_p^0, \mathcal{H}_p^0$
 - 3: **while** $t < t_f$ **do**
 - 4: **Reset grid quantities:** $m_I \leftarrow 0, (m\mathbf{v})_I \leftarrow \mathbf{0}, \mathbf{f}_I^{\text{ext}} \leftarrow \mathbf{0}, \mathbf{f}_I^{\text{int}} \leftarrow \mathbf{0}, m_I^\phi \leftarrow 0, F_I \leftarrow 0$
 - 5: **Particle $\rightarrow$ grid mapping (P2G)**
 - 6: Nodal mass: $m_I^t = \sum_p N_I(\mathbf{x}_p^t) m_p$
 - 7: Nodal momentum: $(m\mathbf{v})_I^t = \sum_p N_I(\mathbf{x}_p^t) (m\mathbf{v})_p$
 - 8: External force: $\mathbf{f}_I^{\text{ext},t} = \sum_p N_I(\mathbf{x}_p^t) m_p \mathbf{b}(\mathbf{x}_p)$
 - 9: Internal force: $\mathbf{f}_I^{\text{int},t} = - \sum_p V_p^t \boldsymbol{\sigma}_p^t \nabla N_I(\mathbf{x}_p^t) \quad \triangleright \boldsymbol{\sigma}_p^t = g(\phi_p^t) \boldsymbol{\sigma}_{0,p}^t \text{ degraded}$
 - 10: Nodal force: $\mathbf{f}_I^t = \mathbf{f}_I^{\text{ext},t} + \mathbf{f}_I^{\text{int},t}$
 - 11: Pseudo–mass: $m_I^\phi = \sum_p N_I(\mathbf{x}_p^t) V_p^0$
 - 12: Damage force: $F_I = \sum_p \left[g'(\phi_p^t) N_I(\mathbf{x}_p^t) \mathcal{H}_p^t + \frac{G_c}{2c_w \ell} \left(\frac{1}{2} w'(\phi_p^t) N_I(\mathbf{x}_p^t) + \ell^2 \nabla N_I(\mathbf{x}_p^t) \cdot \nabla \phi_p^t \right) \right] V_p^0$
 - 13: **end**
 - 14: **Update momenta:** $(m\mathbf{v})_I^{t+\Delta t} = (m\mathbf{v})_I^t + \mathbf{f}_I^t \Delta t$
 - 15: **Update grid phase field:** $\phi_I^{t+\Delta t} = \phi_I^t - \frac{\Delta t}{\eta m_I^\phi} F_I$
 - 16: **Fix Dirichlet nodes** I : e.g. $(m\mathbf{v})_I^t = \mathbf{0}$ and $(m\mathbf{v})_I^{t+\Delta t} = \mathbf{0}$; prescribe ϕ_I on damage–BC nodes
 - 17: **Grid $\rightarrow$ particle mapping (G2P)**
 - 18: Nodal velocities: $\mathbf{v}_I^t = (m\mathbf{v})_I^t / m_I^t, \quad \mathbf{v}_I^{t+\Delta t} = (m\mathbf{v})_I^{t+\Delta t} / m_I^{t+\Delta t}$
 - 19: Update particle velocities (PIC/FLIP blend):
 $\mathbf{v}_p^{t+\Delta t} = \alpha(\mathbf{v}_p^t + \sum_I N_I(\mathbf{x}_p^t) [\mathbf{v}_I^{t+\Delta t} - \mathbf{v}_I^t]) + (1 - \alpha) \sum_I N_I(\mathbf{x}_p^t) \mathbf{v}_I^{t+\Delta t}$
 - 20: Update particle positions: $\mathbf{x}_p^{t+\Delta t} = \mathbf{x}_p^t + \Delta t \sum_I N_I(\mathbf{x}_p^t) \mathbf{v}_I^{t+\Delta t}$
 - 21: Update particle phase field: $\phi_p^{t+\Delta t} = \sum_I N_I(\mathbf{x}_p^t) \phi_I^{t+\Delta t}$
 - 22: Enforce irreversibility: $\phi_p^{t+\Delta t} \leftarrow \max(\phi_p^t, \phi_p^{t+\Delta t})$
 - 23: Velocity gradient: $\mathbf{L}_p^{t+\Delta t} = \sum_I \nabla N_I(\mathbf{x}_p^t) \mathbf{v}_I^{t+\Delta t}$
 - 24: Deformation gradient: $\mathbf{F}_p^{t+\Delta t} = (\mathbf{I} + \mathbf{L}_p^{t+\Delta t} \Delta t) \mathbf{F}_p^t$
 - 25: Update volume: $V_p^{t+\Delta t} = \det(\mathbf{F}_p^{t+\Delta t}) V_p^0$
 - 26: Rate of deformation: $\mathbf{D}_p^{t+\Delta t} = \frac{1}{2} (\mathbf{L}_p^{t+\Delta t} + (\mathbf{L}_p^{t+\Delta t})^T)$
 - 27: Strain increment: $\Delta \boldsymbol{\varepsilon}_p = \Delta t \mathbf{D}_p^{t+\Delta t}$
 - 28: **Update stress (last):** effective $\boldsymbol{\sigma}_{0,p}^{t+\Delta t} = \boldsymbol{\sigma}_{0,p}^t + \mathbf{C}_0 : \Delta \boldsymbol{\varepsilon}_p$; degraded $\boldsymbol{\sigma}_p^{t+\Delta t} = g(\phi_p^{t+\Delta t}) \boldsymbol{\sigma}_{0,p}^{t+\Delta t}$
 - 29: Update elastic energy and history: $\psi_{0,p}^e = \frac{1}{2} \boldsymbol{\varepsilon}_p : \mathbf{C}_0 : \boldsymbol{\varepsilon}_p, \quad \mathcal{H}_p^{t+\Delta t} = \max(\mathcal{H}_p^t, \psi_{0,p}^e)$
 - 30: **end**
 - 31: **Advance time:** $t \leftarrow t + \Delta t$
 - 32: **end while**
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7.1 Time-step and stability considerations

Two explicit stability restrictions act simultaneously, and the time step must respect the smaller of the two.

Elastodynamic (CFL) limit. The momentum update is conditionally stable and limited by the Courant–Friedrichs–Lewy condition,

$$\Delta t \leq \Delta t_{\text{crit}}^u = \frac{h}{c_d}, \quad c_d = \sqrt{\frac{\lambda + 2\mu}{\rho}}, \quad (39)$$

with h the background cell size and c_d the dilatational wave speed. A safety factor $0 < \text{CFL} \leq 1$ is normally applied, $\Delta t = \text{CFL} \cdot h/c_d$.

Parabolic (diffusion) limit of the phase field. Treating (22) as a diffusion equation with diffusivity $D_\phi = \frac{\ell G_c}{2c_w \eta}$, the forward–Euler update is stable only if

$$\Delta t \leq \Delta t_{\text{crit}}^\phi \sim \frac{h^2}{2n D_\phi} = \frac{c_w \eta h^2}{n \ell G_c}, \quad (40)$$

with n the spatial dimension. Equation (40) shows the role of the viscosity η : a larger η relaxes the phase–field time–step restriction (and brings the response closer to rate–independent fracture), but also slows the rate at which the crack can grow, so that η must be small enough not to artificially retard the physical crack speed. In practice η is chosen so that $\Delta t_{\text{crit}}^\phi \gtrsim \Delta t_{\text{crit}}^u$, i.e. so that the elastodynamic CFL condition governs and the phase field can be advanced with the same Δt as the momentum update,

$$\eta \gtrsim \frac{n \ell G_c}{c_w h^2} \Delta t_{\text{crit}}^u = \frac{n \ell G_c}{c_w h c_d}. \quad (41)$$

Consistency. Because $\omega_{\text{diss}} = \eta \dot{\phi} \rightarrow 0$ as $\dot{\phi} \rightarrow 0$, the stationary solutions of the explicit scheme coincide with the equilibrium solutions of the original quasi–static variational theory: the viscous term affects only the transient path towards equilibrium, not the converged crack pattern, provided η is sufficiently small. This is the sense in which the dissipative regularisation is a purely *algorithmic* device enabling the explicit time march while preserving the underlying fracture physics.

8 Summary

Starting from the variational phase–field fracture theory, a single constitutive enrichment—the dissipative phase–field micro–force $\omega_{\text{diss}} = \eta \dot{\phi}$ —was introduced. This converts the elliptic damage stationarity condition into a parabolic Allen–Cahn evolution law (22), which can be integrated by an explicit forward–Euler update on a lumped pseudo–mass (37). The update

was embedded in the explicit Update–Stress–Last MPM cycle, yielding the fully explicit USL–MPM–PF solution procedure of Algorithm 1, in which displacement and phase field are advanced within a single time loop subject to the combined CFL/diffusion time–step restriction. Both AT1 and AT2 crack density models are supported through the functions $w(\phi)$ and the normalisation c_w , and damage irreversibility is enforced through the monotone history field $\mathcal{H}$.

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